Prime University
Department of Computer Science and Engineering
Computer Programming Club

All Support by:

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BASIC C PROGRAMS

1. Write a C program to print "Hello World".
2. Write a C program to print your name.
3. Write a C program to take input from the user and display it.
4. Write a C program to add two numbers.
5. Write a C program to subtract two numbers.
6. Write a C program to multiply two numbers.
7. Write a C program to divide two numbers.
8. Write a C program to find the remainder of two numbers.
9. Write a C program to swap two numbers using a third variable.
10. Write a C program to swap two numbers without using a third variable.
11. Write a C program to calculate the average of three numbers.
12. Write a C program to calculate the percentage of five subjects.
13. Write a C program to convert Celsius to Fahrenheit.
14. Write a C program to convert Fahrenheit to Celsius.
15. Write a C program to calculate the area of a circle.
16. Write a C program to calculate the area of a triangle.
17. Write a C program to calculate the area of a rectangle.
18. Write a C program to calculate the perimeter of a rectangle.
19. Write a C program to calculate simple interest.
20. Write a C program to calculate compound interest.
21. Write a C program to convert seconds into hours, minutes and seconds.
22. Write a C program to convert days into years, months and days.
23. Write a C program to calculate gross salary.
24. Write a C program to calculate electricity bill.

DECISION-MAKING PROGRAMS

25. Write a C program to check whether a number is positive, negative or zero.
26. Write a C program to check whether a number is even or odd.
27. Write a C program to check whether a number is divisible by 5.
28. Write a C program to check whether a number is divisible by both 3 and 5.
29. Write a C program to find the greater between two numbers.
30. Write a C program to find the smaller between two numbers.
31. Write a C program to check whether a student has passed or failed.
32. Write a C program to check whether a person is eligible to vote.
33. Write a C program to check whether a year is a leap year.
34. Write a C program to check whether a character is a vowel or consonant.
35. Write a C program to find the largest among three numbers.
36. Write a C program to find the smallest among three numbers.
37. Write a C program to find the largest among four numbers.
38. Write a C program to find the smallest among four numbers.
39. Write a C program to determine the grade of a student based on marks.
40. Write a C program to determine whether a triangle is equilateral, isosceles or scalene.
41. Write a C program to calculate income tax based on salary.
42. Write a C program to check admission eligibility based on marks.
43. Write a C program to create a simple calculator using switch-case.
44. Write a C program to display the day of the week using switch-case.
45. Write a C program to display the month name using switch-case.
46. Write a C program to display the number of days in a month using switch-case.
47. Write a C program to check whether a character is a vowel using switch-case.
48. Write a C program to convert a digit into its word using switch-case.

LOOP PROGRAMS

49. Write a C program to print numbers from 1 to N.
50. Write a C program to print numbers from N to 1.
51. Write a C program to print all even numbers from 1 to N.
52. Write a C program to print all odd numbers from 1 to N.
53. Write a C program to print the multiplication table of a number.
54. Write a C program to find the sum of numbers from 1 to N.
55. Write a C program to find the sum of all even numbers from 1 to N.
56. Write a C program to find the sum of all odd numbers from 1 to N.
57. Write a C program to calculate the factorial of a number.
58. Write a C program to count the number of digits in a number.
59. Write a C program to find the sum of digits of a number.
60. Write a C program to find the product of digits of a number.
61. Write a C program to reverse a number.
62. Write a C program to check whether a number is a palindrome.
63. Write a C program to check whether a number is prime.
64. Write a C program to print all prime numbers from 1 to N and count number.

65. Write a C program to generate Fibonacci series.
66. Write a C program to find the power of a number using a loop.
67. Write a C program to find the GCD of two numbers.
68. Write a C program to find the LCM of two numbers.
69. Write a C program to check whether a number is an Armstrong number.
70. Write a C program to print Armstrong numbers within a range.
71. Write a C program to check whether a number is a perfect number.
72. Write a C program to print perfect numbers within a range.
73. Write a C program to check whether a number is a strong number.
74. Write a C program to print strong numbers within a range.

PATTERN PROGRAMS

75. Write a C program to print a right-aligned triangle pattern.
76. Write a C program to print a pyramid pattern.
77. Write a C program to print an inverted pyramid pattern.
78. Write a C program to print the following number pattern:
79. Write a C program to print Floyd's triangle.
80. Write a C program to print Pascal's triangle.
81. Write a C program to print a diamond pattern.
82. Write a C program to print a hollow square pattern.
83. Write a C program to print a hollow triangle pattern.

ARRAY PROGRAMS

84. Write a C program to input and display elements of an array.
85. Write a C program to find the sum of array elements.
86. Write a C program to find the average of array elements.
87. Write a C program to find the maximum element of an array.
88. Write a C program to find the minimum element of an array.
89. Write a C program to count even and odd elements in an array.
90. Write a C program to count positive and negative elements in an array.
91. Write a C program to reverse an array.
92. Write a C program to copy one array into another array.
93. Write a C program to merge two arrays.
94. Write a C program to find duplicate elements in an array.
95. Write a C program to remove duplicate elements from an array.
96. Write a C program to find the frequency of each element in an array.
97. Write a C program to find the second largest element in an array.
98. Write a C program to find the second smallest element in an array.
99. Write a C program to separate even and odd elements of an array.
100. Write a C program to sort an array in ascending order.
101. Write a C program to sort an array in descending order.
102. Write a C program to insert an element into an array.
103. Write a C program to delete an element from an array.
104. Write a C program to perform linear search.

105. Write a C program to perform binary search.
106. Write a C program to count the occurrence of an element in an array.
107. Write a C program to find the first occurrence of an element.
108. Write a C program to find the last occurrence of an element.

MATRIX PROGRAMS

109. Write a C program to input and display a matrix.
110. Write a C program to add two matrices.
111. Write a C program to subtract two matrices.
112. Write a C program to multiply two matrices.
113. Write a C program to find the transpose of a matrix.
114. Write a C program to find the sum of all elements of a matrix.
115. Write a C program to find the maximum element of a matrix.
116. Write a C program to find the minimum element of a matrix.
117. Write a C program to find the sum of the main diagonal.
118. Write a C program to find the sum of the secondary diagonal.
119. Write a C program to check whether a matrix is an identity matrix.
120. Write a C program to check whether a matrix is symmetric.
121. Write a C program to check whether a matrix is upper triangular.
122. Write a C program to check whether a matrix is lower triangular.
123. Write a C program to count zero and non-zero elements of a matrix.
124. Write a C program to find the sum of each row of a matrix.
125. Write a C program to find the sum of each column of a matrix.

STRING PROGRAMS

126. Write a C program to input and display a string.
127. Write a C program to find the length of a string.
128. Write a C program to copy one string into another string.
129. Write a C program to compare two strings.
130. Write a C program to concatenate two strings.
131. Write a C program to reverse a string.
132. Write a C program to check whether a string is a palindrome.
133. Write a C program to count vowels in a string.
134. Write a C program to count consonants in a string.
135. Write a C program to count digits in a string.
136. Write a C program to count spaces in a string.
137. Write a C program to count words in a string.
138. Write a C program to count uppercase letters in a string.
139. Write a C program to count lowercase letters in a string.
140. Write a C program to convert lowercase letters into uppercase.
141. Write a C program to convert uppercase letters into lowercase.
142. Write a C program to toggle the case of characters in a string.
143. Write a C program to find the frequency of each character in a string.

144. Write a C program to find duplicate characters in a string.
145. Write a C program to remove spaces from a string.
146. Write a C program to remove duplicate characters from a string.
147. Write a C program to find the first occurrence of a character.
148. Write a C program to find the last occurrence of a character.
149. Write a C program to search for a substring.
150. Write a C program to replace a character in a string.
151. Write a C program to remove a character from a string.
152. Write a C program to sort characters of a string.
153. Write a C program to check whether two strings are anagrams.
154. Write a C program to find the largest word in a sentence.
155. Write a C program to find the smallest word in a sentence.

FUNCTION PROGRAMS

156. Write a C program to add two numbers using a function.
157. Write a C program to find the maximum of two numbers using a function.
158. Write a C program to find the minimum of two numbers using a function.
159. Write a C program to calculate factorial using a function.
160. Write a C program to check prime number using a function.
161. Write a C program to check palindrome using a function.
162. Write a C program to check Armstrong number using a function.
163. Write a C program to generate Fibonacci series using a function.
164. Write a C program to find GCD using a function.
165. Write a C program to find LCM using a function.
166. Write a C program to calculate power using a function.
167. Write a C program to find the sum of array elements using a function.
168. Write a C program to find the maximum element of an array using a function.
169. Write a C program to find string length using a function.
170. Write a C program to reverse a string using a function.
171. Write a C program using a function with no argument and no return value.
172. Write a C program using a function with arguments and no return value.
173. Write a C program using a function with no arguments and a return value.
174. Write a C program using a function with arguments and a return value.

POINTER PROGRAMS

175. Write a C program to demonstrate a basic pointer.
176. Write a C program to access a variable using a pointer.
177. Write a C program to swap two numbers using pointers.
178. Write a C program to add two numbers using pointers.
179. Write a C program to find the maximum of two numbers using pointers.
180. Write a C program to find the minimum of two numbers using pointers.
181. Write a C program to reverse an array using pointers.
182. Write a C program to find string length using a pointer.

183. Write a C program to copy a string using a pointer.
184. Write a C program to reverse a string using a pointer.
185. Write a C program to compare two strings using pointers.
186. Write a C program to demonstrate pointer to pointer.
187. Write a C program to allocate memory dynamically using malloc().
188. Write a C program to allocate memory dynamically using calloc().
189. Write a C program to resize dynamically allocated memory using realloc().
190. Write a C program to release dynamically allocated memory using free().

MATHEMATICAL PROGRAMS

191. Write a C program to check even or odd.
192. Write a C program to check prime number.
193. Write a C program to calculate factorial.
194. Write a C program to generate Fibonacci series.
195. Write a C program to calculate GCD.
196. Write a C program to calculate LCM.
197. Write a C program to check palindrome number.
198. Write a C program to check Armstrong number.
199. Write a C program to check strong number.
200. Write a C program to check perfect number.
201. Write a C program to check automorphic number.
202. Write a C program to check neon number.
203. Write a C program to check Harshad number.
204. Write a C program to check spy number.
205. Write a C program to check happy number.
206. Write a C program to check Disarium number.
207. Write a C program to check whether a number is a perfect square.
208. Write a C program to calculate the power of a number.
209. Write a C program to calculate square root.
210. Write a C program to find the sum of natural numbers.
211. Write a C program to find the sum of squares of natural numbers.
212. Write a C program to find the sum of cubes of natural numbers.
213. Write a C program to count the digits of a number.
214. Write a C program to reverse the digits of a number.
215. Write a C program to find the sum of digits.
216. Write a C program to find the product of digits.

SEARCHING PROGRAMS

217. Write a C program to perform linear search.
218. Write a C program to perform binary search.

SORTING PROGRAMS

219. Write a C program to sort an array using Bubble Sort.

220. Write a C program to sort an array using Selection Sort.
221. Write a C program to sort an array using Insertion Sort.
222. Write a C program to sort an array using Merge Sort.
223. Write a C program to sort an array using Quick Sort.
224. Write a C program to sort an array in ascending order.
225. Write a C program to sort an array in descending order.
226. Write a C program to sort even and odd numbers separately.
227. Write a C program to sort a list of strings alphabetically.

STRUCTURE AND UNION

228. Write a C program to create and display a structure.
229. Write a C program to store and display student information using structure.
230. Write a C program to store employee information using structure.
231. Write a C program to store book information using structure.

RECURSION PROGRAMS

232. Write a C program to calculate factorial using recursion.
233. Write a C program to generate Fibonacci series using recursion.
234. Write a C program to find the sum of natural numbers using recursion.
235. Write a C program to find the sum of digits using recursion.
236. Write a C program to reverse a number using recursion.
237. Write a C program to find GCD using recursion.
238. Write a C program to calculate power using recursion.
239. Write a C program to perform binary search using recursion.
240. Write a C program to solve Tower of Hanoi using recursion.

FILE HANDLING PROGRAMS

241. Write a C program to create a file.
242. Write a C program to write data into a file.
243. Write a C program to read data from a file.
244. Write a C program to append data to a file.
245. Write a C program to count characters in a file.
246. Write a C program to count words in a file.
247. Write a C program to count lines in a file.
248. Write a C program to copy data from one file to another.
249. Write a C program to search for a word in a file.
250. Write a C program to store student records in a file.